MATH 128a: Homework #8

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Due by: **Wednesday, October 29, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from the textbook, chapter 3.1

Solve problems 1bc, 3bc, 6b, 10, 13bc of Section 3.1 of the textbook.

(If the problem has a different number in the 9th edition book, it is in parentheses)

Answer of Problem 1

Exercise 3.1.1. For the given functions $f(x)$, let $x_0 = 0,\ x_1 = 0.6$ and $x_2 = 0.9$. Construct interpolation polynomials of degree at most one and at most two to approximate $f(0.45)$ and find the absolute error.

(b) $f(x) = \sqrt{1+x}$.

(c) $f(x) = \ln(x+1)$

Solution.

- **Degree 1 (Linear) interpolation:**

$$p_1(x) = f(x_0)\frac{x-x_1}{x_0-x_1} + f(x_1)\frac{x-x_0}{x_1-x_0}.$$

- **Degree 2 (Quadratic) interpolation:**

$$p_2(x) = f(x_0)\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} + f(x_1)\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} + f(x_2)\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}.$$

For $x_0 = 0,\ x_1 = 0.6,\ x_2 = 0.9$ and $x = 0.45$:

$$L_0(0.45) = 0.125, \qquad L_1(0.45) = 1.125, \qquad L_2(0.45) = -0.25.$$

(b) $f(x) = \sqrt{1+x}$.

$$f(0) = 1, \quad f(0.6) = 1.2649110641, \quad f(0.9) = 1.3784048752, \quad f(0.45) = 1.2041594579.$$

Linear interpolation:

$$p_1(0.45) = 1 + \frac{1.2649110641 - 1}{0.6} \cdot 0.45 = 1.1986832981,$$

$$|f(0.45) - p_1(0.45)| = 0.0054761598.$$

Quadratic interpolation:

$$p_2(0.45) = 1(0.125) + 1.2649110641(1.125) + 1.3784048752(-0.25) = 1.2034237283,$$

$$|f(0.45) - p_2(0.45)| = 0.0007357296.$$

(c) $f(x) = \ln(1+x)$.

$$f(0) = 0, \quad f(0.6) = 0.4700036292, \quad f(0.9) = 0.6418538862, \quad f(0.45) = 0.3715635564.$$

Linear interpolation:

$$p_1(0.45) = \frac{0.4700036292}{0.6} \cdot 0.45 = 0.3525027219,$$

$$|f(0.45) - p_1(0.45)| = 0.0190608345.$$

Quadratic interpolation:

$$p_2(0.45) = 0(0.125) + 0.4700036292(1.125) + 0.6418538862(-0.25) = 0.3682906114,$$

$$|f(0.45) - p_2(0.45)| = 0.0032729451.$$

Exercise 3.1.3. Use Theorem 3.3 to find an error bound for the approximations in Exercise 1.

Solution.
If P_n is the interpolating polynomial of degree n at nodes $x_0, \ldots, x_n$, then for each x there exists ξ between the nodes with

$$f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)(x - x_1)\cdots(x - x_n),$$

hence the uniform bound

$$|f(x) - P_n(x)| \le \frac{M_{n+1}}{(n+1)!}\left|(x - x_0)(x - x_1)\cdots(x - x_n)\right|,$$

where $M_{n+1} = \max_{t\in[I]} |f^{(n+1)}(t)|$ on an interval I containing the nodes and x.
Use $x_0 = 0,\ x_1 = 0.6,\ x_2 = 0.9$ and $x = 0.45$. Note

$$|(0.45 - 0)(0.45 - 0.6)| = 0.45 \cdot 0.15 = 0.0675, \qquad |(0.45 - 0)(0.45 - 0.6)(0.45 - 0.9)| = 0.030375.$$

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(b) $f(x) = \sqrt{1+x}$.

Compute derivatives and maxima on $[0, 0.9]$:

$$f''(x) = -\frac{1}{4(1+x)^{3/2}}, \quad |f''(x)| \le M_2 = \frac{1}{4} \quad (\text{max at } x = 0),$$

$$f^{(3)}(x) = \frac{3}{8(1+x)^{5/2}}, \quad |f^{(3)}(x)| \le M_3 = \frac{3}{8}.$$

Thus the error bounds at $x = 0.45$ are

$$|f(0.45) - p_1(0.45)| \le \frac{M_2}{2!}\, 0.0675 = \frac{1/4}{2} \cdot 0.0675 = \frac{1}{8} \cdot 0.0675 = 0.0084375,$$

$$|f(0.45) - p_2(0.45)| \le \frac{M_3}{3!}\, 0.030375 = \frac{3/8}{6} \cdot 0.030375 = \frac{1}{16} \cdot 0.030375 = 0.0018984375.$$

(c) $f(x) = \ln(1+x)$.

Compute derivatives and maxima on $[0, 0.9]$:

$$f''(x) = -\frac{1}{(1+x)^2}, \quad |f''(x)| \le M_2 = 1 \quad (\text{max at } x = 0),$$

$$f^{(3)}(x) = \frac{2}{(1+x)^3}, \quad |f^{(3)}(x)| \le M_3 = 2.$$

Thus the error bounds at $x = 0.45$ are

$$|f(0.45) - p_1(0.45)| \le \frac{M_2}{2!}\, 0.0675 = \frac{1}{2} \cdot 0.0675 = 0.03375,$$

$$|f(0.45) - p_2(0.45)| \le \frac{M_3}{3!}\, 0.030375 = \frac{2}{6} \cdot 0.030375 = \tfrac{1}{3} \cdot 0.030375 = 0.010125.$$

Exercise 3.1.6. Use appropriate Lagrange interpolating polynomials of degrees one, two, and three to approximate each of the following:

(b) $f(0)$ if $f(-0.5) = 1.93750,\ f(-0.25) = 1.33203,\ f(0.25) = 0.800781,\ f(0.5) = 0.687500$

Solution.

(b) We use the Lagrange formulas

$$p_n(x) = \sum_{i=0}^{n} f(x_i) \prod_{\substack{j=0 \\ j \ne i}}^{n} \frac{x - x_j}{x_i - x_j},$$

and evaluate at $x = 0$.

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Degree 1. Choose the two nearest nodes -0.25, 0.25. Then

$$p_1(0) = f(-0.25)\frac{0-0.25}{-0.25-0.25} + f(0.25)\frac{0-(-0.25)}{0.25-(-0.25)} = \frac{f(-0.25)+f(0.25)}{2}.$$

Hence

$$p_1(0) = \frac{1.33203 + 0.800781}{2} = 1.0664055.$$

Degree 2. Take the three nearest nodes -0.5, -0.25, 0.25. Evaluate the Lagrange basis at 0:

$$L_{-0.5}(0) = \frac{(0+0.25)(0-0.25)}{(-0.5+0.25)(-0.5-0.25)} = \frac{0.25\cdot(-0.25)}{(-0.25)(-0.75)} = \frac{-0.0625}{0.1875} = -\tfrac{1}{3},$$

$$L_{-0.25}(0) = \frac{(0+0.5)(0-0.25)}{(-0.25+0.5)(-0.25-0.25)} = \frac{0.5\cdot(-0.25)}{0.25\cdot(-0.5)} = \frac{-0.125}{-0.125} = 1,$$

$$L_{0.25}(0) = \frac{(0+0.5)(0+0.25)}{(0.25+0.5)(0.25+0.25)} = \frac{0.5\cdot 0.25}{0.75\cdot 0.5} = \frac{0.125}{0.375} = \tfrac{1}{3}.$$

Thus

$$p_2(0) = f(-0.5)\left(-\tfrac{1}{3}\right) + f(-0.25)(1) + f(0.25)\left(\tfrac{1}{3}\right).$$

Numerically,

$$p_2(0) = 1.93750\cdot\left(-\tfrac{1}{3}\right) + 1.33203\cdot 1 + 0.800781\cdot\left(\tfrac{1}{3}\right) \approx 0.9531236667.$$

Degree 3. Use all four nodes $-0.5, -0.25, 0.25, 0.5$. The Lagrange coefficients at 0 are

$$L_{-0.5}(0) = \frac{(0+0.25)(0-0.25)(0-0.5)}{(-0.5+0.25)(-0.5-0.25)(-0.5-0.5)} = \tfrac{1}{8},$$

$$L_{-0.25}(0) = \frac{(0+0.5)(0-0.25)(0-0.5)}{(-0.25+0.5)(-0.25-0.25)(-0.25-0.5)} = -\tfrac{9}{32},$$

$$L_{0.25}(0) = \frac{(0+0.5)(0+0.25)(0-0.5)}{(0.25+0.5)(0.25+0.25)(0.25-0.5)} = \tfrac{9}{32},$$

$$L_{0.5}(0) = \frac{(0+0.5)(0+0.25)(0-0.25)}{(0.5+0.25)(0.5+0.5)(0.5-0.25)} = -\tfrac{1}{8}.$$

(These four coefficients sum to 1.) Hence

$$\begin{aligned} p_3(0) &= \tfrac{1}{8}f(-0.5) - \tfrac{9}{32}f(-0.25) + \tfrac{9}{32}f(0.25) - \tfrac{1}{8}f(0.5) \\ &= \tfrac{1}{8}\cdot 1.93750 - \tfrac{9}{32}\cdot 1.33203 + \tfrac{9}{32}\cdot 0.800781 - \tfrac{1}{8}\cdot 0.6875 \\ &\approx 0.984374. \end{aligned}$$

Results (rounded):

$$p_1(0) \approx 1.066406, \qquad p_2(0) \approx 0.953124, \qquad p_3(0) \approx 0.984374.$$

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Exercise 3.1.10. Let $f(x) = \sqrt{x - x^2}$ and $P_2(x)$ be the interpolation polynomial on $x_0 = 0$, x_1, and $x_2 = 1$. Find the largest value of x_1 in $(0, 1)$ for which $f(0.5) - P_2(0.5) = -0.25$.

Solution. Since $f(x_0) = 0$ and $f(x_2) = 0$, the interpolating polynomial $P_2(x)$ is $f(x_0)(\cdots) + f(x_1) \cdot \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} + f(x_2)(\cdots) = 0 + (\sqrt{x_1 - x_1^2})\frac{x(x-1)}{x_1(x_1-1)} + 0$.
We need to solve the equation $f(0.5) - P_2(0.5) = -0.25$ for x_1.

$$\begin{aligned}
f(0.5) - P_2(0.5) &= -0.25 \\
\sqrt{0.5 - 0.5^2} - \sqrt{x_1 - x_1^2}\frac{(0.5)(0.5-1)}{x_1(x_1-1)} &= -0.25 \\
\sqrt{0.25} - \frac{\sqrt{x_1 - x_1^2}}{x_1^2 - x_1} \cdot -0.25 &= -0.25 \\
0.5 - 0.25 \cdot \frac{1}{\sqrt{x_1 - x_1^2}} &= -0.25 \\
-0.25 \cdot \frac{1}{\sqrt{x_1 - x_1^2}} &= -0.75 \\
\frac{1}{\sqrt{x_1 - x_1^2}} &= 3 \\
x_1 - x_1^2 &= \frac{1}{9} \\
x_1^2 - x_1 + \frac{1}{9} &= 0
\end{aligned}$$

Using the quadratic formula and simplifying gives $x_1 = \frac{3 \pm \sqrt{5}}{6}$. In decimal form, we have $x_1 = 0.872678$ or $x_1 = 0.127322$. The largest value of x_1 that satisfies the required condition is thus 0.872678 or $\frac{3+\sqrt{5}}{6}$.

Exercise 3.1.13. Construct the Lagrange interpolating polynomials for the following functions and find a bound for the absolute error on the interval $[x_0, x_n]$.

(b) $f(x) = \sin(\ln x)$, $x_0 = 2.0$, $x_1 = 2.4$, $x_2 = 2.6$, $n = 2$

(c) $f(x) = \ln x, x_0 = 1$, $x_1 = 1.1$, $x_2 = 1.3$, $x_3 = 1.4$, $n = 3$

Solution.

(b) $f(x) = \sin(\ln x)$, $x_0 = 2.0$, $x_1 = 2.4$, $x_2 = 2.6$ (degree $n = 2$).

Lagrange form (degree 2):

$$p_2(x) = \sum_{i=0}^{2} f(x_i)\, L_i(x), \qquad L_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j},$$

with

$$f(2.0) = \sin(\ln 2) = 0.6389612763, \quad f(2.4) = 0.7678438771, \quad f(2.6) = 0.8166090488.$$

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Expanded form (for convenience):

$$p_2(x) = -0.130634405510282\,x^2 + 0.896997886150849\,x - 0.632496873946955.$$

Error bound. By Theorem 3.3,

$$f(x) - p_2(x) = \frac{f^{(3)}(\xi)}{3!}(x - x_0)(x - x_1)(x - x_2)$$

for some $\xi \in [2.0, 2.6]$. Compute

$$f^{(3)}(x) = \frac{3\sin(\ln x) + \cos(\ln x)}{x^3},$$

so $|f^{(3)}(x)| \le \frac{4}{x^3}$ and on $[2, 2.6]$ we may take $M_3 = \max_{[2,2.6]} |f^{(3)}(x)| \le 4/2^3 = 0.5$. Hence for all $x \in [2, 2.6]$,

$$|f(x) - p_2(x)| \le \frac{M_3}{6} \max_{x\in[2,2.6]} |(x - 2.0)(x - 2.4)(x - 2.6)|.$$

$\max_{[2,2.6]} |(x - 2)(x - 2.4)(x - 2.6)| \le 0.6 \cdot 0.4 \cdot 0.6 = 0.144$, so

$$|f(x) - p_2(x)| \le \frac{0.5}{6} \cdot 0.144 = 0.012.$$

(c) $f(x) = \ln x,\ x_0 = 1,\ x_1 = 1.1,\ x_2 = 1.3,\ x_3 = 1.4$ (degree $n = 3$).
Lagrange form (degree 3):

$$p_3(x) = \sum_{i=0}^{3} f(x_i)\, L_i(x), \qquad f(x_i) = \ln(x_i).$$

Values:

$$f(1) = 0, \quad f(1.1) = \ln(1.1), \quad f(1.3) = \ln(1.3), \quad f(1.4) = \ln(1.4).$$

Expanded form:

$$p_3(x) = 0.197005607906711\,x^3 - 1.06259031597423\,x^2 + 2.53245289941788\,x - 1.66686819135037.$$

Error bound. For $n = 3$ Theorem 3.3 gives

$$f(x) - p_3(x) = \frac{f^{(4)}(\xi)}{4!}(x - 1)(x - 1.1)(x - 1.3)(x - 1.4).$$

For $\ln x$ we have $f^{(4)}(x) = -6/x^4$, so $|f^{(4)}(x)| = 6/x^4 \le 6$ on $[1, 1.4]$. Thus $M_4 = 6$ and

$$|f(x) - p_3(x)| \le \frac{M_4}{4!} \max_{x\in[1,1.4]} |(x - 1)(x - 1.1)(x - 1.3)(x - 1.4)|.$$

The product is maximized at $x = 1.2$ with value 0.0004, hence

$$|f(x) - p_3(x)| \le \frac{6}{24} \cdot 0.0004 = 0.25 \cdot 0.0004 = 1.0 \times 10^{-4}.$$

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Problem 2: Problems from the textbook, chapter 3.2

Solve problems 3, 5 of Section 3.2 of the textbook.

(If the problem has a different number in the 9th edition book, it is in parentheses)

Answer of Problem 2

Exercise 3.2.3. Use Neville's method to approximate $\sqrt{3}$ with the following functions and values.

(a) $f(x) = 3^x$ and the values $x_0 = -2$, $x_1 = -1$, $x_2 = 0$, $x_3 = 1$, and $x_4 = 2$.

(b) $f(x) = \sqrt{x}$ and the values $x_0 = 0$, $x_1 = 1$, $x_2 = 2$, $x_3 = 4$, and $x_4 = 5$.

(c) Compare the accuracy of the approximation in parts (a) and (b).

Solution.
We use Neville's recursion

$$P_{i,0} = f(x_i), \qquad P_{i,j} = \frac{(x - x_{i+j})\,P_{i,j-1} + (x_i - x)\,P_{i+1,j-1}}{x_i - x_{i+j}}, \quad j = 1, 2, \ldots$$

evaluated at the target x; the degree-n interpolant value is $P_{0,n}$.

(a) Take $f(x) = 3^x$ and evaluate at $x = \frac{1}{2}$ so that $f(\frac{1}{2}) = \sqrt{3}$. With nodes $x_i = -2, -1, 0, 1, 2$, the Neville tableau (rounded) is

i	$P_{i,0}$	$P_{i,1}$	$P_{i,2}$	$P_{i,3}$	$P_{i,4}$
0	0.111111	0.666667	1.500000	1.777778	1.708333
1	0.333333	1.333333	1.833333	1.666667	
2	1.000000	2.000000	1.500000		
3	3.000000	0.000000			
4	9.000000				

Hence the 4th-degree interpolation gives

$$\sqrt{3} \approx P_{0,4} = 1.7083333333333335,$$

with absolute error $|\sqrt{3} - P_{0,4}| \approx 2.372 \times 10^{-2}$.

(b) Take $f(x) = \sqrt{x}$ and evaluate at $x = 3$ so that $f(3) = \sqrt{3}$. With nodes $x_i = 0, 1, 2, 4, 5$, the Neville tableau (rounded) is

i	$P_{i,0}$	$P_{i,1}$	$P_{i,2}$	$P_{i,3}$	$P_{i,4}$
0	0.000000	3.000000	1.242641	1.621320	1.690607
1	1.000000	1.828427	1.747547	1.736798	
2	1.414214	1.707107	1.726049		
3	2.000000	1.763932			
4	2.236068				

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Thus

$$\sqrt{3} \approx P_{0,4} = 1.6906067646231164,$$

with absolute error $|\sqrt{3} - P_{0,4}| \approx 4.144 \times 10^{-2}$.

(c) Part (a) is more accurate: $|1.7083333 - \sqrt{3}| \approx 2.37 \cdot 10^{-2}$ versus $|1.6906068 - \sqrt{3}| \approx 4.14 \cdot 10^{-2}$ in part (b). Intuitively, evaluating 3^x at $x = \frac{1}{2}$ with symmetric nearby nodes (0 and 1) yields a tighter fit here than interpolating $\sqrt{x}$ at $x = 3$ using the given, less symmetric grid.

Exercise 3.2.5. Neville's method is used to approximate $f(0.4)$, giving the following table:

$x_0 = 0$	$P_0 = 1$			
$x_1 = 0.25$	$P_1 = 2$	$P_{01} = 2.6$		
$x_2 = 0.5$	P_2	P_{12}	P_{012}	
$x_3 = 0.75$	$P_3 = 8$	$P_{23} = 2.4$	$P_{123} = 2.96$	$P_{0123} = 3.016$

Determine $P_2 = f(0.5)$.

Solution. We use the equation $P_{23} = \frac{(x-x_2)P_3 - (x-x_3)P_2}{x_3 - x_2}$. Solving for P_2 gives

$$\begin{aligned} P_2 &= \frac{(x - x_2)P_3 - (x_3 - x_2)P_{23}}{x - x_3} \\ &= \frac{(0.4 - 0.5) \cdot 8 - (0.75 - 0.5) \cdot 2.4}{0.4 - 0.75} = 4 \end{aligned}$$

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